East Genomics

BRCA1, BRCA2 & PALB2 Gene Testing

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BRCA1, BRCA2 and PALB2 gene testing

For most people, cancer happens by chance. In a small number of people with breast cancer (about 5%), it happens because they have a change in a gene, called a variant, in BRCA1, BRCA2, or PALB2.

Some treatments, like PARP inhibitors, work best for people who have a gene change in BRCA1 or BRCA2. A genetic test can check these genes to help decide if this treatment may work for you.

Gene changes in BRCA1, BRCA2, and PALB2 can increase the chance of developing several types of cancer:

- **Breast cancer in both men and women.**
- **Ovarian cancer.** BRCA1 and BRCA2 gene changes increase this risk. PALB2 only raises this risk a little.

- **Prostate cancer.** This is linked mainly to BRCA2 gene changes.
- **Pancreatic cancer**

These changes are more common in people who have had both breast and ovarian cancer, certain types of these cancers, or a strong family history of them.

Why am I being offered this test?

You are being offered a test to look for gene changes that may have been the cause of your cancer diagnosis.

What are the benefits to me?

Knowing if you have a change in the BRCA1, BRCA2, or PALB2 genes can help your cancer team understand more about your cancer.

This can show if certain treatments, called PARP inhibitors, might work for you and can also help when planning surgery.

Genetic test results can also tell you and your family about your chances of getting cancer in the future. If you have a change in one of these genes, you can talk about ways to lower the risk, like extra screening or surgery if needed.

What is a gene?

Our genes are like a set of instructions inside our body that tell it how to grow, work, and look. Each gene gives instructions for one specific job, like what colour our eyes are or how our body breaks down food.

We have two copies of most of our genes. These come from our parents - one copy from our mother and one copy from our father. That's why we often look like our parents.

Sometimes, a gene can have a change (called a variant) that stops it working properly. If that change causes a health problem, it's called a

pathogenic variant. Changes in some genes can increase the chance of developing cancer.

What results might I get?

1. No gene change found. This is the most common result. It means we didn't find a genetic reason for your breast cancer. This can be reassuring, but sometimes there may still be a genetic cause that we don't yet know about. Your care team might still offer you a Genetics clinic appointment if needed, if you haven't already seen them.

2. A gene change. This means there is a genetic reason for your breast cancer. Your care team will use this information to help plan your treatment. You will be offered an appointment with the Genetics team to talk more about what this means. This is an opportunity to discuss what the result means for your health and whether genetic testing should be offered to other family

members. You will be able to ask any other questions at this appointment too.

3. A gene change is found, but we're not sure what it means. This is rare (in less than 1 in 100 people). Sometimes we find a change in a gene, but we don't know yet if it causes cancer. We call this a variant of unknown significance or VUS. You'll be offered an appointment with the Genetics team to talk about this in more detail.

Do I have to have the test?

No, it's your choice. The test can help guide your treatment, but if you decide not to have it, your doctors will still give you the best care possible.

It would be helpful to complete the test soon, as the results may influence your treatment options. If there is a delay, the results may not be available in time to guide your care, and national guidelines mean the test might not be possible later on. If at any point you decide you would

like to go ahead, please speak to your hospital team.

What if I am not sure if I want to have the test?

If you haven't already seen the Genetics team, you can ask to have an appointment with them. They can help you think through the pros and cons of testing.

What will happen next if I say yes?

You'll be asked to sign a form to show you agree to the test. You'll also be asked about your choices for research. A blood sample will be taken for the test.

How will I receive the results of the test?

The team who arranged your test will give you the results, usually in about 8 weeks.

They'll set up a time to talk with you about what the results mean. If you haven't seen the Genetics team you may also have an appointment with them to talk about what the results mean for you and your family.

Will my information be kept private?

Yes. Your information will be kept safe and private. Only your care team and a few trained staff will see your details.

Everyone involved must follow strict rules to protect your privacy. Some limited information is shared with the National Disease Registration Service.

Are any other genetic tests available?

There are other gene changes that can also increase the chance of getting breast cancer. However, these gene changes are either a less common cause of breast cancer, or do not affect whether someone can have a PARP inhibitor. Because of this, they are not included in this test.

If people in your family (blood relatives) have had breast, ovarian, prostate, or pancreatic cancer, childhood cancers, or sarcoma (a cancer of tissues like bone), you might need more detailed genetic tests.

You may also be offered extra tests if you have Jewish heritage, a grandparent from Westray (Orkney) or Whalsay (Shetland), were diagnosed with breast cancer at a very young age, have had other cancers, or health problems like thyroid lumps, bowel polyps, or harmless skin growths.

Ask your healthcare practitioner if you think you might need more genetic tests.

About this leaflet

This leaflet provides general information only. Your care will be based on your own needs and the services available in your area. This leaflet does not replace guidance from your healthcare professional.

Need more help?

If you want more information or need help understanding this leaflet, please contact your healthcare team.

Your feedback

We would value your feedback on this leaflet, whether you are a patient, carer, family member or NHS clinician. Please visit https://bit.ly/leaflet-feedback to provide your feedback.

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